

TOURS VAL DE LOIRE AP, France

WMO: 072400

Lat: 47.4444N

Lon: 0.7275E

Elev: 357

StdP: 14.51

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.3	27.1	17.7	13.6	26.7	21.5	16.4	29.4	27.1	51.7	24.0	50.2	7.3	50	0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	89.1	68.5	85.1	67.0	81.6	65.9	70.7	84.3	68.8	81.3	67.2	78.5	7.7	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	98.3	75.6	64.6	92.3	73.3	63.0	87.1	71.5	34.7	84.6	33.1	81.3	31.8	78.6	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.1	18.5	16.4	DB	20.1	95.2	4.8	3.5	16.6	97.7	13.8	99.8	11.2	101.8	7.7	104.3	(4)
			WB	19.3	73.9	4.9	2.2	15.8	75.4	12.9	76.7	10.2	77.9	6.7	79.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.8	40.9	42.1	47.0	51.4	57.7	64.2	68.1	67.8	61.7	55.4	46.7	41.7
	DBStd	11.53	7.09	6.74	5.99	5.88	5.70	6.16	5.49	5.63	5.32	6.15	6.32	7.13
	HDD50	1121	287	226	129	51	6	0	0	0	22	134	266	
	HDD65	4433	747	641	556	409	234	88	29	31	127	300	549	722
	CDD50	2511	5	5	38	93	246	426	562	551	351	190	35	9
	CDD65	347	0	0	0	1	9	63	126	117	28	3	0	0
	CDH74	3371	0	0	0	21	116	644	1199	1103	267	21	0	0
	CDH80	1109	0	0	0	1	15	208	411	413	60	1	0	0
Wind	WSAvg	8.2	9.2	9.1	9.0	8.5	8.0	7.6	7.5	7.1	7.4	7.6	8.2	9.0
Precipitation	PrecAvg	26.5	2.5	2.1	2.0	2.0	2.4	2.0	1.9	1.8	2.1	2.5	2.8	2.5
	PrecMax	36.6	5.4	5.7	4.8	5.7	5.4	5.3	4.4	6.7	6.9	6.0	6.6	5.0
	PrecMin	20.4	0.3	0.2	0.1	0.3	0.4	0.0	0.2	0.0	0.4	0.1	0.5	0.5
	PrecStd	4.2	1.3	1.3	1.1	1.2	1.3	1.2	1.1	1.3	1.4	1.5	1.5	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.4	61.1	69.5	77.0	81.9	92.7	93.8	95.8	86.1	76.4	64.0	57.6
		MCWB	53.2	51.5	55.5	60.9	66.0	71.2	68.8	71.2	65.5	63.6	58.2	54.3
	2%	DB	54.5	56.3	64.3	71.5	77.1	85.5	88.6	89.3	80.8	71.5	60.5	55.6
		MCWB	51.9	49.9	53.6	58.5	63.4	67.7	68.4	68.4	64.6	61.4	56.5	52.7
	5%	DB	52.6	53.5	60.2	66.8	73.4	81.2	84.6	84.3	76.6	67.5	58.0	53.8
		MCWB	50.1	49.5	51.9	55.8	61.8	66.4	66.6	67.2	62.8	60.2	54.7	51.5
	10%	DB	50.4	51.4	57.0	62.4	69.5	76.8	80.9	80.1	72.5	64.5	55.6	51.9
		MCWB	48.1	48.3	50.8	53.4	59.7	64.5	65.5	65.4	61.0	58.9	52.7	49.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.4	53.6	57.3	62.8	68.2	72.9	72.2	73.2	68.7	65.2	59.9	55.6
		MCDB	55.6	58.1	66.2	74.1	79.4	86.3	87.1	88.4	79.5	73.0	62.3	56.9
	2%	WB	52.4	51.6	54.9	59.3	65.1	69.9	69.9	70.6	66.4	63.0	57.1	53.4
		MCDB	54.2	54.3	61.3	69.5	74.3	82.2	84.1	84.4	76.5	68.8	60.1	55.2
	5%	WB	50.4	50.1	53.2	56.6	62.6	67.6	68.2	68.5	64.4	61.4	54.9	51.8
		MCDB	52.2	52.6	58.4	64.9	71.7	78.2	81.2	80.9	74.0	66.4	57.4	53.6
	10%	WB	48.4	48.5	51.6	54.3	60.4	65.5	66.6	66.7	62.6	59.7	53.1	50.0
		MCDB	50.2	51.3	56.1	61.2	68.0	75.4	78.1	77.2	70.2	63.8	55.3	51.7
Mean Daily Temperature Range	5% DB	MDBR	9.1	11.8	15.1	17.1	17.5	19.1	19.8	19.9	18.7	14.5	10.9	9.4
		MCDBR	10.0	14.6	20.8	24.0	24.0	25.8	26.2	27.0	24.7	18.5	12.5	10.3
		MCWBR	8.2	10.2	12.3	13.0	12.7	11.5	9.8	10.7	10.9	10.3	8.9	8.5
	5% WB	MCDBR	9.3	12.5	16.9	21.6	21.8	23.1	22.6	22.7	20.8	16.2	11.7	9.5
		MCWBR	8.2	9.7	10.7	12.2	12.1	11.3	9.4	10.1	10.3	9.8	8.9	8.3
Clear-Sky Solar Irradiance	taub	0.322	0.342	0.370	0.396	0.398	0.411	0.403	0.392	0.365	0.361	0.334	0.318	
	taud	2.445	2.386	2.309	2.238	2.277	2.278	2.295	2.344	2.400	2.421	2.443	2.425	
	Ebn at Noon	240	259	268	271	275	271	271	270	267	248	233	226	
	Edh at Noon	21	28	35	41	41	41	40	36	31	26	21	19	
All-Sky Solar Radiation	RadAvg	330	593	978	1412	1662	1865	1808	1591	1232	736	398	282	
	RadStd	31	81	120	171	150	181	114	148	87	63	42	41	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page